A novel guidance algorithm and comparison of nonlinear control strategies applied to an indoor quadrotor

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***Abstract*— In this paper we propose a cascaded approach for the full control of a quadrotor operating in indoor spaces. Sliding mode controller with a saturation function is applied to achieve robust attitude tracking performance without control chattering. A novel guidance algorithm and a feedback-linearization based position controller is proposed, which ensures that the MAV follows the shortest path between the waypoints. Also, we discuss the on-board implementation of state-estimation and 3D localization techniques for an MAV. The robustness and effectiveness of the proposed control law are validated through simulations and experiments under external disturbances. Further, the test-bench results are compared with those obtained from PI-PID and backstepping control.**

***Index Terms*— Quadrotor, Nonlinear control, Sliding mode, Feedback linearization, Indoor navigation, Sensor fusion**

I. INTRODUCTION

Autonomous aerial robots are increasingly used for outdoor applications such as photography, surveillance, fault detection and delivery. However, indoor environments poses interesting challenges such as unreliability of GPS, lack of free-space for navigation, presence of obstacles. This calls for alternate methods for localization, aggressive maneuvers as well as obstacle avoidance, etc. The aim of this paper is to implement and evaluate the performance of various classes of controllers and discuss localization techniques for indoor navigation of vertical take-off and landing micro aerial vehicles, MAVs.

Due to its simple design with fewer mechanical complexities, quadrotors serve as a great platform for research in control and automation of indoor MAVs. On the flip side, quadrotors are under-actuated systems having highly nonlinear and coupled dynamics. Many control strategies have been developed for similar class of under-actuated systems and extended to the case of quadrotors [1]. In the relevant literature of quadcopter control, the main focus has been in the following areas: (a) Linear - PID, LQR [3] (b) Nonlinear - backstepping, sliding mode control [5], [7] (c) Optimal - NMPC [8] and (d) Intelligent - Fuzzy logic [9]. Among them, sliding mode control has been an efficient control algorithm that can be implemented on embedded boards and can handle large uncertainties, bounded external disturbances and non-linearities [7].

A cascaded control algorithm is proposed, with its novelties being a smooth saturation integral sliding mode attitude control, and an intuitive radial co-ordinates based guidance algorithm with feedback-linearization position control. Furthermore, this work contributes to the literature of experimental comparison of different classes of controllers - PI-PID, backstepping and SMC, under external wind disturbances. State estimation and localization are an imperative aspect of a closed loop control system. In this paper, we present a low cost alternative to commercial autopilots such as Pixhawk [10] and discuss the relevance of ultra-wideband based triangulation [14].

The remainder of this paper is organized as follows: Section II describes the modeling of the system dynamics. The cascaded controller design is proposed in section III and simulation results are presented. Section IV provides the description of the experimental setup, while Section V discusses about state estimation. Finally, the experimental comparisons between the controllers are presented in Section VI and conclusions are stated in Section VII.

II. DYNAMICS MODELLING

Let us consider earth fixed frame E and body fixed frame B, as seen in Fig. 1. Position, $\boldsymbol{\xi} = [x, y, z]^T$ and the attitude, $\boldsymbol{\eta} = [\phi, \theta, \psi]^T$ of the quadcopter are defined in the inertial frame, E. The dynamic model is derived using Newton-Euler formalism [2]. Dynamics of any rigid body under the influence of external forces, $\boldsymbol{F}$ and moments, $\boldsymbol{\tau}$ expressed in a rotating frame of reference (body axes) is,

$$\begin{bmatrix} mI_{3\times3} & 0 \\ 0 & J \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{v}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\omega} \times m\boldsymbol{v} \\ \boldsymbol{\omega} \times J\boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} \boldsymbol{F} \\ \boldsymbol{\tau} \end{bmatrix} \tag{1}$$

Where, m is the mass, J the inertia matrix, $\boldsymbol{v}$ the body linear velocity and $\boldsymbol{\omega}$ is body angular rates. In the inertial frame, the equations can be written as,

$$\begin{aligned} \dot{\boldsymbol{\xi}} &= \boldsymbol{v} \\ m\dot{\boldsymbol{v}} &= R\boldsymbol{F} \\ \dot{\boldsymbol{\eta}} &= W_\eta \boldsymbol{\omega} \\ J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau} \end{aligned} \tag{2}$$

Using 3-2-1 Euler angles rotation, the rotation matrix,

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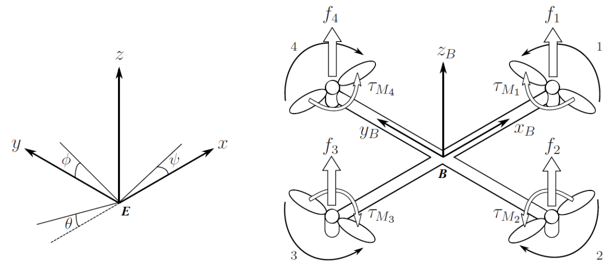

Fig. 1. Quadcopter Model [2] and reference frames

$R \in SO(3)$ from the body frame to the inertial frame is constructed as shown in (3).

$$R = \begin{bmatrix} C_\psi C_\theta & C_\psi S_\theta S_\phi - S_\psi S_\phi & C_\psi S_\theta C_\phi + S_\psi S_\phi \\ S_\psi C_\theta & S_\psi S_\theta S_\phi + C_\psi C_\phi & S_\psi S_\theta C_\phi - C_\psi S_\phi \\ -S_\theta & C_\theta S_\phi & C_\theta C_\phi \end{bmatrix} \quad (3)$$

The transformation matrix for angular velocities, W_η from the body frame to inertial frame is,

$$W_\eta = \begin{bmatrix} 1 & S_\phi T_\theta & C_\phi T_\theta \\ 0 & C_\phi & -S_\phi \\ 0 & S_\phi / C_\theta & C_\phi / C_\theta \end{bmatrix} \quad (4)$$

where, $S_x = \sin x$, $C_x = \cos x$ and $T_x = \tan x$. Quadcopters are highly dynamic systems, thus an appropriate model should also include the gyroscopic effects resulting from the four rotors. Therefore, the dynamics are formulated as,

$$\begin{aligned} \dot{\boldsymbol{\xi}} &= \boldsymbol{v} \\ m\dot{\boldsymbol{v}} &= -g\boldsymbol{e_3} + R\boldsymbol{e_3}T \\ \dot{\boldsymbol{\eta}} &= W_\eta \boldsymbol{\omega} \\ J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} - J_r(\boldsymbol{\omega} \times \boldsymbol{e_3})\Omega + \boldsymbol{M} \end{aligned} \quad (5)$$

$\boldsymbol{e_3} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$ and g is acceleration due to gravity. Thrust, T and Moment, $\boldsymbol{M}$ are defined as,

$$\begin{aligned} T &= k \sum_{i=1}^{4} \Omega_i^2 \\ \boldsymbol{M} = \begin{bmatrix} M_\phi \\ M_\theta \\ M_\psi \end{bmatrix} &= \begin{bmatrix} kl(\Omega_4^2 - \Omega_2^2) \\ kl(\Omega_3^2 - \Omega_1^2) \\ d(\Omega_2^2 + \Omega_4^2 - \Omega_1^2 - \Omega_3^2) \end{bmatrix} \end{aligned} \quad (6)$$

where, J_r is rotor inertia, Ω_i the rotor speed, and Ω is the net rotor speed defined in (8). k is the thrust co-efficient, d the counter-moment drag co-efficient and l is the arm length.

III. CONTROL LAW

In order to derive the control law, we assume that the Euler angles are small. Also, since the quadcopter is a perfect cross configuration, the cross moment of inertia terms are supposed negligible. Then, the model can be written in a state-space form, $\dot{\boldsymbol{X}} = \boldsymbol{f}(\boldsymbol{X}, \boldsymbol{U})$, by introducing state vector, $\boldsymbol{X} = [x_1 \ x_2 \ . \ . \ . \ x_{12}]^T \in \Re^{12}$ and input vector, $\boldsymbol{U} = [u_1 \ u_2 \ u_3 \ u_4]^T \in \Re^4$ defined as,

$$\boldsymbol{X} = [\boldsymbol{\eta} \ \ \dot{\boldsymbol{\eta}} \ \ \boldsymbol{\xi} \ \ \dot{\boldsymbol{\xi}}]^T \qquad \boldsymbol{U} = [M_\phi \ \ M_\theta \ \ M_\psi \ \ T]^T$$

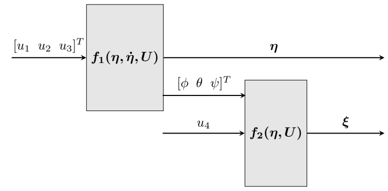

Fig. 2. Rotational and translational sub-systems

$$\dot{\boldsymbol{X}} = \boldsymbol{f}(\boldsymbol{X}, \boldsymbol{U}) = \begin{pmatrix} x_4 \\ x_5 \\ x_6 \\ a_1 x_5 x_6 - a_2 x_5 \Omega + b_1 u_1 \\ a_3 x_6 x_4 + a_4 x_4 \Omega + b_2 u_2 \\ a_5 x_4 x_5 + b_3 u_3 \\ x_{10} \\ x_{11} \\ x_{12} \\ b_4 w_1 \\ b_4 w_2 \\ -g + b_4 w_3 \end{pmatrix} \quad (7)$$

The constants and Ω are defined as,

$$\begin{array}{l|l} a_1 = (J_y - J_z)/J_x & b_1 = 1/J_x \\ a_2 = J_r/J_x & b_2 = 1/J_y \\ a_3 = (J_z - J_x)/J_y & b_3 = 1/J_z \\ a_4 = J_r/J_y & b_4 = 1/m \\ a_5 = (J_x - J_y)/J_z & \Omega = \Omega_1 + \Omega_3 - \Omega_2 - \Omega_4 \end{array} \quad (8)$$

The transformed control input, $\boldsymbol{w}$ for the position dynamics is defined in (9),

$$\begin{aligned} w_1 &= (\cos x_3 \sin x_2 \cos x_1 + \sin x_3 \sin x_1) u_4 \\ w_2 &= (\sin x_3 \sin x_2 \cos x_1 - \cos x_3 \sin x_1) u_4 \\ w_3 &= (\cos x_2 \cos x_1) u_4 \end{aligned} \quad (9)$$

In the approximated model, the angle dynamics do not depend on the translational states. Furthermore, the position dynamics depend only on the angles. Therefore, the complete system (7) can be divided into 2 sub-systems as illustrated in Fig. 2. From the open-loop behavior of the system, it is noted that the angular rotation dynamics are relatively much faster than the translational dynamics. Hence, a cascaded control approach is proposed as depicted in Fig. 3. The outer loop

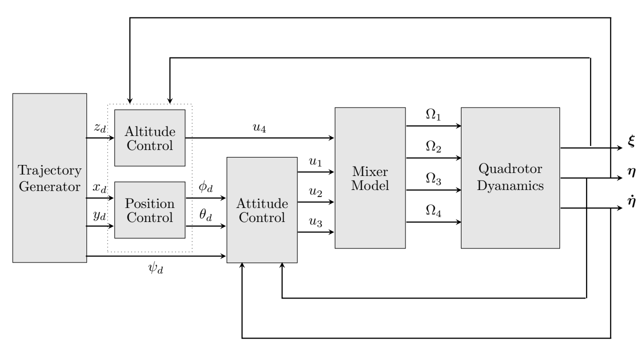

Fig. 3. Closed loop cascaded control schematic

controls the translational motion of the quadcopter to track the desired trajectory. This controller generates the control input u_4 and the desired attitude (ϕ_d, θ_d) of the quadcopter, while the desired heading angle, ψ_d is 0. The inner loop controls the rotational motion of the quadcopter to track the desired attitude. This loop produces control inputs u_1, u_2 and u_3. Lastly, the mixer model calculates the requisite rotor speeds by inverting the functions described in (6).

A. Sliding Mode Control for Attitude Tracking

Firstly, we consider the trajectory tracking error of roll, x_1 and later extend the result to the other angles.

$$\tilde{x}_1 = x_1 - x_{1d} \tag{10}$$

The sliding surface is defined as

$$s_1 = \dot{\tilde{x}}_1 + 2\lambda_1 \tilde{x}_1 + \lambda_1^2 \int_0^t \tilde{x}_1(\tau) d\tau \tag{11}$$

The sliding condition (12) is defined such that $s_1 \dot{s}_1 < 0$ and hence, the trajectory converges to $s_1 = 0$ plane.

$$\frac{1}{2}\frac{d}{dt}{s_1}^2 \leq -\eta_1 s_1 sat(s_1) \tag{12}$$

$sat(.)$ is defined as,

$$sat(s) = \begin{cases} s/\mu & \text{for } s \in [-\mu, \mu] \\ sign(s) & \text{otherwise} \end{cases} \tag{13}$$

Where, λ_1, η_1 and μ are strictly positive constants. Assuming $\ddot{x}_{1d} = 0$, from the system dynamics (7) and differentiating (11) once,

$$\begin{aligned} \dot{s}_1 &= \dot{x}_4 + 2\lambda_1 \dot{\tilde{x}}_1 + \lambda_1^2 \tilde{x}_1 \\ &= a_1 x_5 x_6 - a_2 x_5 \Omega + b_1 u_1 + 2\lambda_1 \dot{\tilde{x}}_1 + \lambda_1^2 \tilde{x}_1 \end{aligned} \tag{14}$$

The control law is derived by enforcing condition (12)

$$u_1 = \frac{1}{b_1}(-a_1 x_5 x_6 + a_2 x_5 \Omega - 2\lambda_1 \dot{\tilde{x}}_1 - \lambda_1^2 \tilde{x}_1 - k_1 sat(s_1)) \tag{15}$$

With, $k_1 = F_1 + \eta_1$ being the robustness parameter. Where, F_1 is the absolute bound on the unmodeled effects in the roll dynamics. Similarly, the other control inputs are derived.

$$\begin{aligned} u_2 &= \frac{1}{b_2}(-a_3 x_6 x_4 - a_4 x_4 \Omega - 2\lambda_2 \dot{\tilde{x}}_2 - \lambda_2^2 \tilde{x}_2 - k_2 sat(s_2)) \\ u_3 &= \frac{1}{b_3}(-a_5 x_4 x_5 - 2\lambda_3 \dot{\tilde{x}}_3 - \lambda_3^2 \tilde{x}_3 - k_3 sat(s_3)) \end{aligned} \tag{16}$$

with,

$$\begin{aligned} \tilde{x}_2 &= x_2 - x_{2d} \\ s_2 &= \dot{\tilde{x}}_2 + 2\lambda_2 \tilde{x}_2 + \lambda_2^2 \int_0^t \tilde{x}_2(\tau) d\tau \\ \tilde{x}_3 &= x_3 - x_{3d} \\ s_3 &= \dot{\tilde{x}}_3 + 2\lambda_3 \tilde{x}_3 + \lambda_3^2 \int_0^t \tilde{x}_3(\tau) d\tau \end{aligned} \tag{17}$$

The global stability of the control law is guaranteed by considering the candidate Lyapunov function,

$$V_i = \frac{1}{2} s_i^2 \qquad \dot{V}_i = s_i \dot{s}_i \tag{18}$$

Since, the control law is derived by enforcing the constraint (12), $\dot{V}_i$ is negative definite. V_i is a positive definite function and hence, from LaSalle's theorem, trajectories tend to the invariant set, $s_i = 0$. Therefore, the overall stability of the rotational subsystem is guaranteed [6]. It is worthwhile to note that adopting a *saturation* function has manifold advantages. Primarily, usage of a saturation function smoothens out the control input and hence, it greatly reduces the control chattering phenomenon that is an inherent characteristic of a classical switching sliding mode controller. Secondly, even in the event of external disturbances, the control law (15) & (16), ensures that the trajectories tend to a boundary layer with thickness 2μ about $s = 0$. According to [6], error bounds on s translates to bounds on error in states as,

$$|s_1| \leqslant \mu_1 \implies |\tilde{x}_1| \leqslant \frac{\mu_1}{\lambda_1} \text{ and } |\dot{\tilde{x}}_1| \leqslant 2\mu_1 \tag{19}$$

Lastly, introduction of an integral term in the sliding surface reduces the steady state error and improves the tracking performance. The above remarks are corroborated by the results reported in the later sections, Figs. 10 - 12.

B. Analytical Approach to Tuning the Parameters

The proposed controller contains 9 parameters, $(\lambda_1, \lambda_2, \lambda_3, \mu_1, \mu_2, \mu_3, k_1, k_2, k_3)$. The equations are rewritten so as to provide an intuitive insight into tuning the parameters. Within the boundary layer, the closed loop dynamics under the saturation control law behaves as a PID control system. With, $e_1 = x_{1d} - x_1$,

$$u_1 = \frac{1}{b_1}\begin{pmatrix} [2\lambda_1 + k_1/\mu_1]\,\dot{e}_1 + \left[\lambda_1^2 + 2\lambda_1 k_1/\mu_1\right] e_1 \\ + \left[\lambda_1^2\right] \int_0^t e_1(\tau) d\tau - [\text{system dynamics}] \end{pmatrix} \tag{20}$$

Comparing this with a conventional PID controller, the proportional and derivative gains are "$\lambda_i^2 + 2\lambda_i(k_i/\mu_i)$" and "$2\lambda_i + (k_i/\mu_i)$" respectively, with "$\lambda_i^2$" being the integral gain. Outside the boundary layer, the proposed control law acts as a robust PD controller with $K_p = \lambda_i^2$ and $K_d = 2\lambda_i$, and the constant, k_i providing the robustness to the controller. However, this analysis just provides a rough, intuitive approach to tuning parameters. Hence, the parameters were tuned simultaneously using the Nonlinear Control Design blockset (NCD) in Simulink.

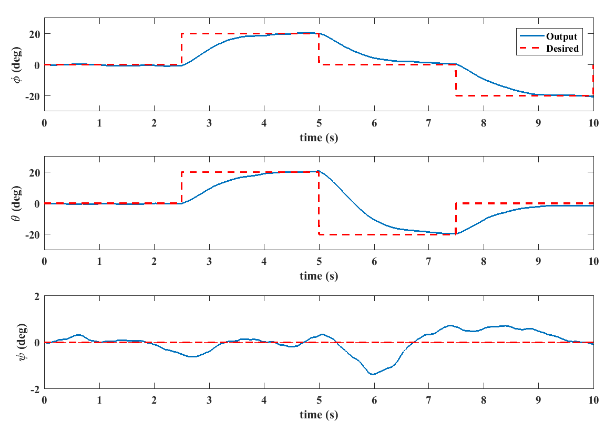

Fig. 4. Simulation: Attitude tracking using *saturation*-integral SMC

C. Guidance & Feedback Linearisation based Translational Control

The input to the linear translational subsystem is transformed as in (9), so as to implement feedback linearisation based control. This allows us to extend linear system analysis for the controller design of each of the three translational DOF.

1) Altitude, z control: A PD control law, w_3 is implemented to hold the quadcopter at a given height, h. Transforming back as described in (9), results in a non-linear controller defined as,

$$u_4 = \frac{1}{b_4 \cos\phi \cos\theta}(g + k_{p,z}e_z + k_{d,z}\dot{e}_z) \tag{21}$$

where, $e_z = h - z(t)$

2) Position, (x, y) control: Applying a decoupled controller [2] to x and y separately, will not result in an optimal trajectory as illustrated in Fig. 6. Therefore, this calls for an alternate algorithm to ensure that quadcopter follows the shortest path to the desired setpoint. The following proposed algorithm utilizes a radial co-ordinates based guidance and a linear PD controller in these transformed co-ordinates. In the radial co-ordinates, Fig. 5, the errors are defined as,

$$\begin{aligned} e_r &= \sqrt{e_x^2 + e_y^2} \\ v_n &= -\dot{x}\sin\beta + \dot{y}\cos\beta \end{aligned} \tag{22}$$

where,

$$\begin{aligned} e_x = x_d - x \qquad\qquad & e_y = y_d - y \\ \beta = \tan^{-1}&\left(\frac{e_y}{e_x}\right) \end{aligned} \tag{23}$$

The controller is derived such that $e_r \rightarrow 0$ and $v_n \rightarrow 0$. This ensures that the quadcopter reaches the desired setpoint in the shortest path. The first condition gives the tilt of the thrust vector, α off the vertical axis. The second condition gives the correction, β_{corr} to be given to the projection of the thrust vector on the (x, y) plane. Mathematically,

$$\begin{aligned} \alpha &= k_{p,r}e_r + k_{d,r}\dot{e}_r \\ \beta_{corr} &= -k_{p,v}v_n - k_{d,v}\dot{v}_n \end{aligned} \tag{24}$$

Such a radial co-ordinates based guidance law, results in the following accelerations in the (x, y) plane. From (7), the transformed control input, w_1 and w_2 are:

$$\begin{aligned} \ddot{x} &= b_4 w_1 = b_4 u_4 \sin\alpha\cos\gamma \\ \ddot{y} &= b_4 w_2 = b_4 u_4 \sin\alpha\sin\gamma \\ \gamma &= \beta + \beta_{corr} \end{aligned} \tag{25}$$

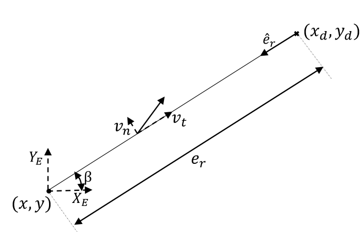

Fig. 5. Radial co-ordinate axes

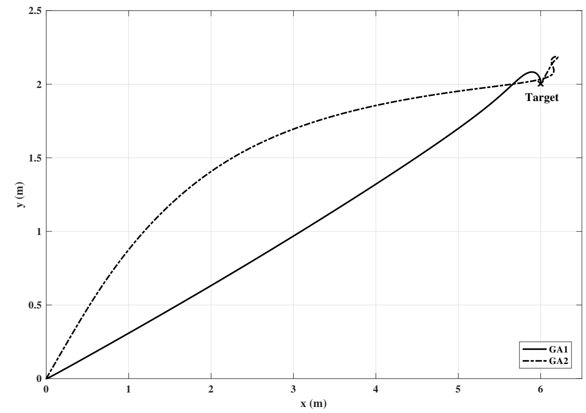

Fig. 6. Simulation: Comparison between 2 guidance algorithm GA1 - radial co-ordinates based guidance algorithm and control; GA2 - decoupled (x, y) control

From (9), solving for the desired attitude, ϕ_d and θ_d,

$$\begin{bmatrix} \sin\phi_d \\ \tan\theta_d \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \sin\alpha\cos\gamma \\ \tan\alpha\sin\gamma \end{bmatrix} \tag{26}$$

The error dynamics, e_r and v_n are second order and first order, respectively. Hence, the gains are analytically tuned using linear analysis - pole placement. The trajectory generator in Fig. 3 defines the way-points. The target co-ordinates of the next waypoint is updated when the quadcopter enters a sphere centered around this point. The radius of this sphere $(0.1m)$ is the maximum admitted error. Fig. 7 shows a square trajectory of $3m$ side defined by four way-points. The task was to hold the altitude at $3m$ and follow the four way-points of the square.

IV. QUADCOPTER TEST BENCH

A. Quadcopter Design

The quadrotor vehicle consists of a carbon-fiber frame, rotors, avionics and pertinent sensors for automation as shown in Fig. 8. Each rotor comprises of a brushless DC motor, an ESC and a 10” $\times$ 4.5” propeller. The total weight of the quadcopter is $1220g$, composed of a $450g$ frame, $360g$ propulsion system, a $288g$ $4000mAh$ $3S$ LiPo battery and a $122g$ computation unit with sensors and telemetry.

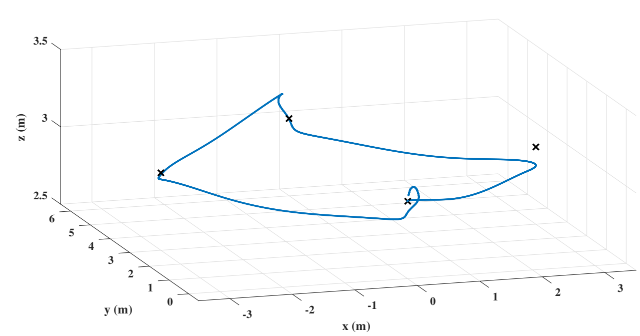

Fig. 7. Simulation: Waypoint tracking with altitude hold

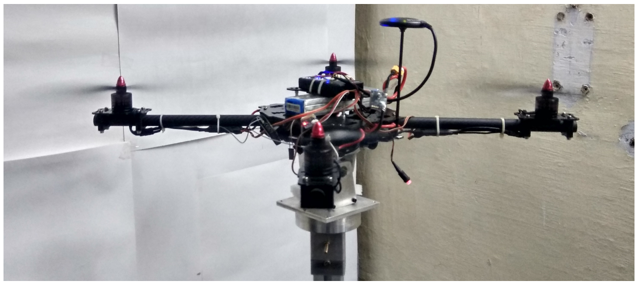

Fig. 8. Quadcopter test-bench

B. Test-bench Design

A test bench is imperative for preliminary analysis of an inherently unstable system. Hence, a 3-DOF test stand was designed and fabricated to evaluate the performance of various attitude controllers. This helps to lock some degrees of freedom, reducing control complexity and avoiding system damage. The stand allows the model to roll or pitch up to $\pm 42°$, while yaw is essentially unconstrained. Further, the support pole was sized based on CFD studies, to minimize the ground effects.

C. Computation Board

To implement the controller, one can modify the firmware of commercial autopilots such as Pixhawk [10]. However, building our own flight controller board provided a low-cost algorithm testing platform. Moreover, this approach provided more flexibility and ensured that we have complete knowledge of the various sub-systems of the quadcopter. The computation unit is based on STM32F446 microcontroller, which has a 180 MHz CPU with 128kB RAM and 512kB flash storage. The board features 8 PWM output pins and several I2C, SPI and UART interfaces for communication with additional sensors.

V. STATE ESTIMATION

A. Attitude Estimation

The attitude feedback is obtained from an inertial sensor, MPU9250 from InvenSense which has a 3-axis gyroscope, accelerometer and compass. The raw data from these sensors have to be fused to obtain the attitude estimate. Various sensor fusion algorithms such as complimentary filter, Madgwick-Mahony filter [11] [12] and Kalman filter were studied. Of all the algorithms, Kalman filter proved to be the computationally least intensive method. The states were considered to be quaternions, which linearizes the prediction dynamics. The attitude of the AV is estimated as mentioned in [13]. Lastly, the quaternions are converted to Euler angles.

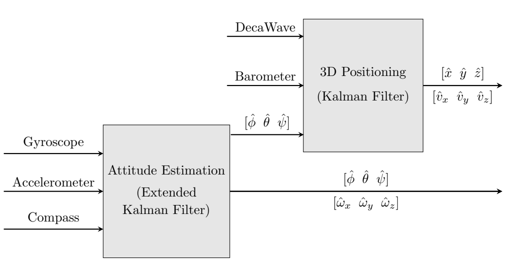

Fig. 9. State estimation schematic

B. 3-D Positioning

The position sensor is based on DW1000 from DecaWave, a ultra-wideband (UWB) transceiver. The positioning setup comprises of 4 anchors in the indoor space and a tag on the quadrotor. Localization with an accuracy of 10cm is achieved through time of flight ranging and particle filter-based triangulation [14]. The accuracy is further enhanced by fusing the inertial data through Kalman filter. The position dynamics are discretized and linearized until there is an explicit presence of input, $\boldsymbol{w}$ term, refer (9).

$$\begin{aligned} \boldsymbol{\xi}_{k+1} &= \boldsymbol{\xi}_k + \dot{\boldsymbol{\xi}}_k T_S + \frac{1}{2}\boldsymbol{w}{T_s}^2 \\ \dot{\boldsymbol{\xi}}_{k+1} &= \dot{\boldsymbol{\xi}}_k + \boldsymbol{w}T_s \end{aligned} \tag{27}$$

Where, T_s is the sampling time of the estimator and $\boldsymbol{w}$ is obtained from angle estimates. System matrices A and B are extracted from (27) with output matrix, $C = I_{6\times 6}$. Subsequently, translational estimates are obtained through the predict-update recursive routine of a Kalman filter. Therefore, through these fusion and filtering algorithms, we get rotational angle estimates with an error range of $\pm 2°$ and translational position feedback is obtained with an absolute maximum error of 5cm.

VI. IMPLEMENTATION ON THE REAL SYSTEM

At present, due to data losses in wireless telemetry and subsequent crashes, it was not possible to release the quadcopter from the test-bench and fly it. Therefore, the implementation of attitude controller alone, has been discussed in this section. The attitude controller and estimation loops run at a frequency of 100Hz and 250Hz, respectively. Other classes of controllers, PI-PID, backstepping [5] and conventional sliding mode control [6], were implemented on the test bench and compared.

A. Steady-state Performance

The 4 controllers are tuned based on step-input responses to achieve the best tracking performance. From Fig. 10, it is observed that $saturation$-integral SMC consistently resulted in least variance of steady state error, whereas $signum$-SMC gave the highest. C1 and C2 gave nearly the same results, largely because of its similar structure as discussed in [4]. The better performance of C4 is mainly attributed to the introduction of an integral term. In the event of external disturbances, variances observed in C1 and C2 increased whereas, C3 and C4 gave almost similar results as without the disturbances, thus affirming the robustness of SMC.

B. Robustness Evaluation

In addition to unmodeled dynamics and system parameter uncertainties, the test bench is subjected to external wind disturbances and robustness of the proposed controller is studied. As seen in Figs. 10, 11, $saturation$-integral SMC

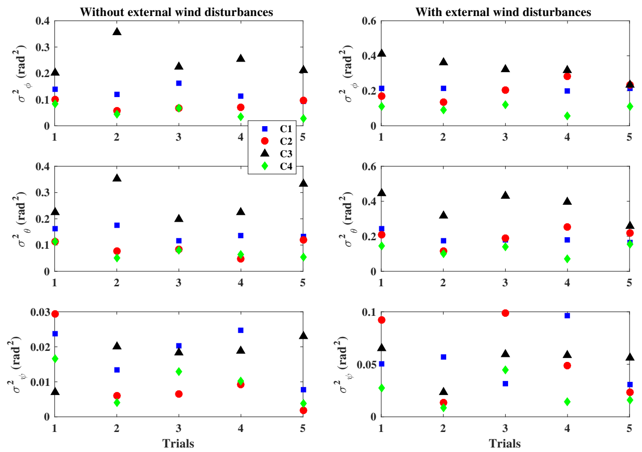

Fig. 10. Experiment: Comparison of variance of steady state error C1 - PI-PID; C2 - Backstepping; C3 - *signum*-SMC; C4 - *saturation*-integral SMC

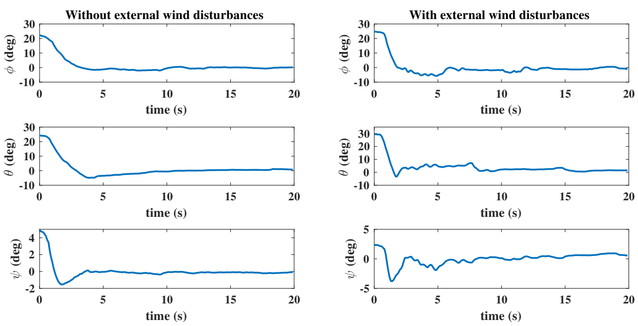

Fig. 11. Experiment: Attitude stabilization using *saturation*-SMC

controller is able to provide robustness to the closed loop system, ensuring better tracking performance.

C. Control Chattering

The relevance of applying a smooth *saturation*-based sliding mode technique over a conventional *signum*-based controller is analyzed from the s-trajectories of the 3 angular rotations. As observed from Fig. 12, control chattering is more evident in the conventional SMC. Therefore, such a switching controller is ill-suited for quadcopter attitude control as it induces high frequency vibrations causing the inertial sensors to drift. Whereas, *saturation*-based SMC gives a smooth control input which is well-adapted to the system dynamics and hence, is more effective.

VII. CONCLUSION

In this paper, we presented a cascaded non-linear control strategy for attitude and position tracking. Various classes of controllers - PI-PID, backstepping and SMC, were compared in different test conditions. As seen from experimental plots, saturation integral sliding mode control provided the best results because of its hybrid formulation. Effectiveness of the proposed guidance and position control algorithm was delineated through simulations. Finally, sensor fusion and localization techniques for an indoor MAV were discussed.

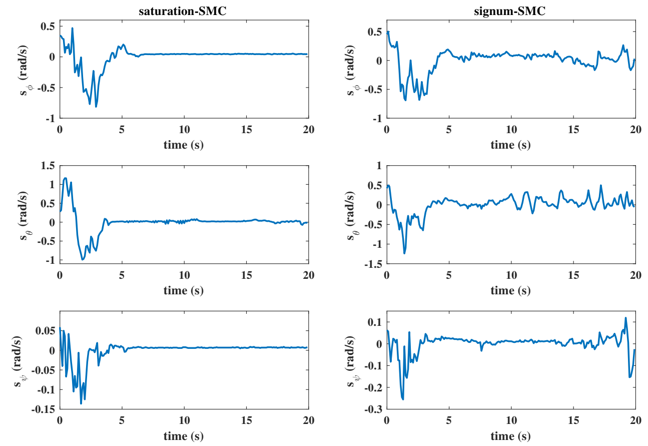

Fig. 12. Experiment: s-trajectories of attitude DOF with *saturation*-SMC and *signum*-SMC

Our future work is to fly the quadrotor with the proposed position controller, thereby achieving autonomous indoor flight.

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